

PROFILE SUMMARY

- Senior Smart Contract Developer with 8 years of experience writing production Solidity contracts and audit deliverables for DeFi protocols on Ethereum mainnet across DeFi lending and AMM protocols, liquid staking systems, and ERC-4626 vault frameworks, specializing in reentrancy-resistant accounting, gas-optimal storage layout, and formal-verification-grade invariants.
- Hands-on coverage across contract language (Solidity 0.8.x), contract framework (Foundry with Forge and Cast), fuzzer (Echidna with Foundry invariant tests), and formal verification (Certora Prover), with upgradeability through UUPS proxy with OpenZeppelin upgrades and credentialed as Code4rena Top 50 wardens list.
- Deep expertise in reentrancy-safe accounting with checks-effects-interactions and pull-payment patterns, gas-optimal storage layout with packed slots, immutables, and bitmap state, proxy upgrade architecture covering UUPS, Transparent, and Diamond storage discipline, and oracle-resilient pricing with TWAPs, sanity bounds, and circuit breakers, applying methodologies such as adversarial-first design where attacker models drive the spec before the first line of contract code and property-based invariants written alongside contracts and proven before audit handoff to deliver DeFi contracts that ship clean audits and survive years on mainnet without an incident.
- Engaged collaborator working cross-functionally with protocol research, external audit firms, and front-end Web3 squads in audit-grade smart contract teams shipping protocol upgrades on a quarterly cadence, contributing to threat-model docs, audit-finding triage, and design reviews for upgradeable contract systems with an ownership-first mindset and clean handoffs.
- Mentor who shares technical excellence and fosters a culture of zero-incident track record on shipped protocol versions and invariant-first authoring that makes the spec executable through PR reviews and threat-model walkthroughs, while running the internal audit-review board and the Solidity hiring loop and authoring widely cited Solidity security write-ups and Certora specs.

TECHNICAL SKILLS

Languages:	Solidity 0.8.x, Vyper 0.3.x, Yul and inline assembly, Rust with Anchor (reading), Move (reading), TypeScript, Python
Frameworks & Toolchains:	Foundry with Forge and Cast, Hardhat, Truffle, Anchor, OpenZeppelin Contracts, Solady, Solmate, Tenderly DevNet
Security & Static Analysis:	Slither with custom detectors, Mythril, Aderyn, Semgrep for Solidity, 4naly3er, checks-effects-interactions, pull-payment, reentrancy guards, access control patterns
Fuzzing & Formal Verification:	Echidna with Foundry invariant tests, Foundry fuzz, Medusa, Halmos, Certora Prover, K Framework, Runtime Verification, property-based testing
Standards & Patterns:	ERC-20, ERC-721, ERC-1155, ERC-4626 vaults, ERC-4337 account abstraction, ERC-2612 permit, EIP-712 typed data, UUPS proxy with OpenZeppelin upgrades, Transparent proxy, Diamond proxy (EIP-2535)
DeFi Primitives & Protocols:	AMM curves (constant product, stableswap), lending and borrowing markets, liquid staking, yield vaults, governance with Compound Governor, Chainlink price feeds, TWAP oracles, flash loans
Chains & Deployment:	Ethereum mainnet, Arbitrum One, Optimism, Base, Polygon zkEVM, zkSync Era, Avalanche C-Chain, Solana mainnet (reading), Etherscan verification, Gnosis Safe multisig, Tenderly DevNet

Certifications & Communities: Code4rena Top 50 wardens list, Sherlock contest watson, Cyfrin Updraft Solidity, Secureum bootcamp graduate, ETHGlobal contributor, Devcon speaker

EDUCATION

National Technical University of Athens [M.Eng. in Electrical and Computer Engineering](#) Athens, Greece • Sep 2013 - Jul 2018

WORK EXPERIENCE

Cyfrin [Senior Smart Contract Developer](#) Athens, Greece • Jun 2021 - Present

- Owned smart contract delivery end to end on **DeFi audit engagements and protocol upgrade reviews** protecting **\$4.6B** in TVL secured, driving **security review, gas tuning, and invariant proofs** across **18** audits with the spec, the implementation, and the proof harness owned by the same engineer.
- Built security and vulnerability prevention into every engagement through **reentrancy guards, checks-effects-interactions, oracle sanity bounds, and access-control matrix reviews**, threat-model docs that mapped each finding to a CWE class, and incident-replay tests that pinned every reported exploit pattern to a passing invariant, surfacing **34** high-severity issues and **71** medium-severity findings across the book.
- Drove gas optimization on hot protocol paths through **packed storage slots, immutables for protocol constants, bitmap accounting, and Yul assembly on hot paths**, a documented gas budget per external call, and Foundry gas snapshots wired into CI to catch regressions, cutting gas per swap from **148k** to **92k** for an aggregate reduction of **38%** across the audited protocols.
- Led audit prep and formal verification with **Slither and Mythril static sweeps, Echidna campaigns on accounting invariants, and Certora Prover specs on access control**, a documented invariant catalog per protocol, and Halmos symbolic runs on price math, proving **62** invariants across the engagement book and putting **10M** fuzz runs through the suite per release.
- Shipped DeFi and protocol logic across **stableswap AMM curves, ERC-4626 vault accounting, Compound-style lending markets, and flash-loan reentrancy hardening**, Chainlink TWAP wrappers with stale-data guards, and a shared liquidation engine reused across three lending protocols, auditing **22** DeFi protocols representing **\$4.6B** in TVL secured at peak.
- Owned testing and test coverage on every contract through **Foundry unit tests, invariant suites, fork tests against mainnet, and adversarial scenarios from documented exploit playbooks**, coverage targets baked into the CI gate, and mutation testing on critical math, hitting **96%** branch coverage and authoring **1.4k** tests across the audited contracts.
- Designed upgradeability and contract architecture around **UUPS proxies with OpenZeppelin upgrades, storage gap discipline, and Diamond facet routing on the larger protocols**, a storage-layout diff tool wired into the audit checklist, and an upgrade-simulation harness that replayed mainnet state against the candidate implementation, shipping **27** proxy upgrades without an incident and catching **6** storage collisions before deployment.

Aave Companies [Solidity Engineer, Protocol](#) Athens, Greece • Aug 2018 - May 2021

- Shipped token and contract standards on the lending protocol through **ERC-20 with ERC-2612 permit, ERC-4626 vault shares, ERC-721 collateral receipts, and EIP-712 typed signatures**, strict interop tests against reference implementations, and clean separation between standard surface and protocol logic, landing **9** standard-compliant contracts behind an aToken book that reached **\$2.1B** in total supply.
- Ran deployment and on-chain operations end to end with **Hardhat scripts gated by Gnosis Safe multisig, Etherscan verification, and Tenderly monitors on every deployment**, a documented runbook per release, and a deployment-replay harness that caught misconfigured parameters before signing, executing **46** mainnet deployments across **7** chains without a failed transaction or a botched verification.
- Held the integration and tooling seat with **typed ABIs with NatSpec, event indexing contracts for The Graph, and audit-firm collaboration on Spearbit and Trail of Bits engagements**, contract-event schemas reused by the frontend, and a triage protocol for external audit findings with a same-day patch SLA, publishing **84** typed ABIs and closing **118** audit findings

across two annual review cycles.

- Anchored core contract development on **lending market core in Solidity 0.8.x on Hardhat, OpenZeppelin Contracts base, and Foundry-driven invariant tests**, a typed math library for index accrual, and a per-asset risk parameter registry, shipping 62 contracts into the protocol that handled **\$3.4B** in TVL at peak.